

Multimodal Large Language Model

Multimodal = Model which can handle more than one data type.

Key Terms in the Language Model Ecosystem ≡

What is an MLLM (Multimodal Large Language Model) or what is Multimodality? - **Clear Definition**

Models examples: Open Source, Close Source, Leaderboards

Why Do LLMs Need Multi Modalities? ≡

What is Multimodal LLM Fine-Tuning? When and Why Should You Use Multimodal Fine-Tuning?

Architecture Overview ≡

What Exactly Gets Fine-Tuned in Multimodal Training?

Dataset Format for Multimodal Fine-Tuning

How to create custom dataset ≡

Normal LLM Fine-Tuning vs Multimodal Fine-Tuning - **Clear Comparison**

Practical Using Unsloth

Key Terms in the Language Model Ecosystem

LLM (Large Language Model)

A Transformer-based model trained on massive datasets to understand and generate language.

Examples: GPTs, Claudes, Geminis, LLaMAs etc...

MLLM (Multimodal Large Language Model)

An LLM extended with additional capability (vision/audio/video).

Examples: GPT-4o, Gemini 2.5, Qwen3-VL, LLaVA etc...

Vision-Language Model (VLM)

Audio-Language Model or Speech Model

SLM (Small Language Model)

A compact language model or Multimodal (typically <10B parameters) optimized for efficiency, speed, and specialization.

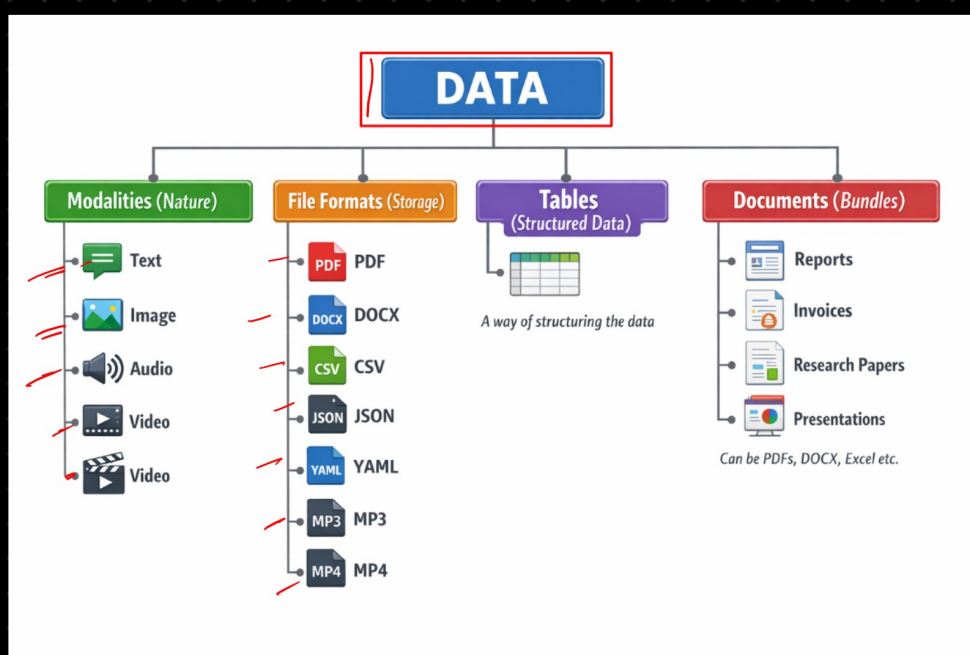
Examples: Phi-3 Mini, Gemma 2B, Gemma Vision, Tiny Llama, Qwen2-VL-2B, LLaVA Small }

Closed-Source

- API-based
- No weight access
- Limited fine-tuning access

Open-Source

- Weight access(download possible)
- Fine-tuning possible
- Local deployment



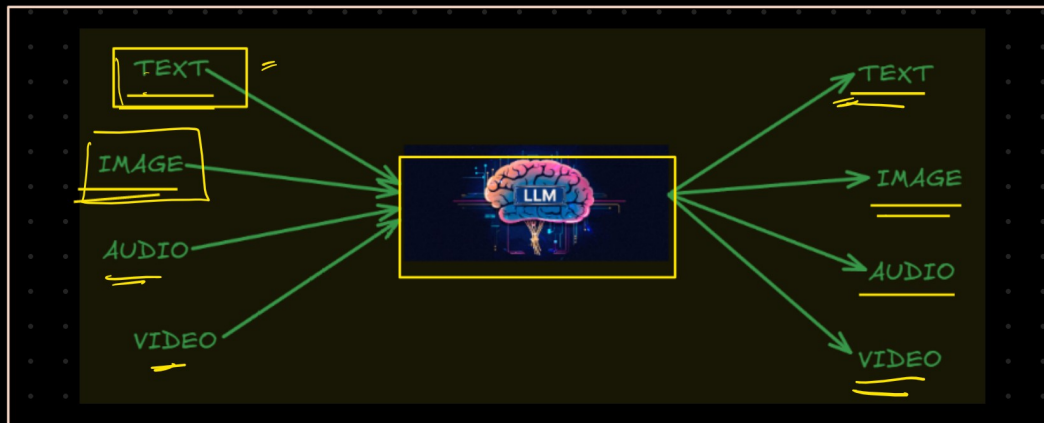
1. What is an MLLM (Multimodal Large Language Model) or what is Multimodality? (Clear Definition)

A multimodal LLM is an AI model that can understand, process and generate multiple types of data (modalities).

Examples of modalities:

-  Text ✓
-  Images ✓
-  Audio ✓
-  Video ✓

Multimodal LLM



Modality Group	Transformation Type	Description / Use Case	Closed-Source Models	Open-Source Models
Text Modality	Text → Text	Classic LLM case.	Open AI GPT models, Anthropic Claude, Google Gemini etc...	Llama, Mistral etc...
Image Modality	Image → Text	Provide an image and extract the content.	Open AI GPT-4o & 5, Google DeepMind Gemini 1.5 / 2.0, Anthropic Claude 3 Vision	Salesforce BLIP-2, Meta LLaVA, Alibaba Group Qwen-VL, Google Gemma Vision, Hugging Face IDEFICS
Image Modality	Text → Image	Generate an image from a text prompt.	OpenAI DALL-E, Midjourney, Gemini-2.5-flash-image, Gemini-3 image preview (Nano banana)	Stability AI: Stable Diffusion, Black Forest Labs: FLUX.x
Image Modality	Image → Image	Modify, enhance, or transform an input image. (Background removal, Style transfer, Super resolution, Face swap etc...)	Adobe Firefly (image editing), Open AI DALL-E (image editing), Nano banana (image editing)	Stability AI: Stable Diffusion (img2img / inpainting)
Audio Modality	Audio → Text	Convert spoken audio into text (Speech Recognition).	OpenAI Whisper API, Google Speech-to-Text	OpenAI Whisper (open model), Mozilla DeepSpeech
Audio Modality	Text → Audio	Convert text into synthesized speech (Text-to-Speech).	OpenAI OpenAI TTS, Google Text-to-Speech	Coqui AI TTS, ESPnet TTS
Audio Modality	Audio → Audio	Transform or modify input audio.	ElevenLabs Voice AI, Adobe Enhance Speech	Meta Voice box
Video Modality	Video = Images + Time + Audio	Video modality combines images + time + audio.	—	—
Video Modality	Text → Video	Generate video from text prompt.	Runway Gen-2, Pika Labs, Open AI SORA	Stability AI Stable Video Diffusion, ModelScope Text-to-Video
Video Modality	Video → Text	Generate description or reasoning from video input. Examples: Video captioning, Scene understanding, Action recognition	Open AI GPT-4o, Gemini Vision Model	Alibaba Group Qwen-VL (video variants), Qwen3-Omni, OpenGVLab InternVideo
Video Modality	Video → Video	Modify or transform existing video. Examples: Style transfer, Video enhancement, Face swap	Runway Video editing tools, Adobe Firefly Video	Stability AI Stable Video Diffusion (img2vid / vid2vid), OpenMMLab / MMEediting

https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard#/

<https://huggingface.co/spaces/lmarena-ai/lmarena-leaderboard>

<https://huggingface.co/papers/trending>

Why Do LLMs Need Multiple Modalities?

Since the Real World Is Multimodal

Humans don't understand the world through text alone.

We use:

- Language (text, speech)
- Audio (tone, emotion)
- Vision (images, video)

and build the Context (environment)

text

AI

HUMAN BRAIN

text (lang.), vision, audio, video

Speak, watch, listen

Models need multiple modalities because the real world is not text-only — intelligence requires seeing, hearing, and understanding context.

Example:

A doctor looks at:

- X-ray image
- Patient report
- Lab values

All together — not just text.

If AI only understands text, it sees only a small part of reality.

What is Multimodal LLM Fine-Tuning?

Adapting a MM Large Language Model to perform a specific task(Domain Specific Task) using multimodal data.
(image + text, audio + text, video + text)

When Should You do Multimodal Fine-Tuning?

Multimodal understanding

OCR reasoning

Visual conversation

Document AI

GPT-4o

Medical analysis

MM → Image, Audio, video

MM → Image

Multimodal LLM fine-tuning is the process of adapting a MM large language model that processes both visual and textual inputs by updating its projection layers and/or language model weights using image-text instruction datasets to improve domain-specific cross-modal reasoning.

ViT: <https://arxiv.org/pdf/2010.11929>

CLIP: <https://arxiv.org/pdf/2103.00020>

Qwen-VL: <https://arxiv.org/pdf/2308.12966>

Qwen-VL-2.5: <https://arxiv.org/pdf/2502.13923>

BLIP-2: <https://arxiv.org/abs/2301.12597>

LLAVA: <https://arxiv.org/abs/2304.08485>

OpenAI Dalle: <https://arxiv.org/pdf/2103.00020>

Google Imagen: <https://arxiv.org/pdf/2205.11487> Nano banner

Architecture overview

Image modality = Image to text, Text to Image

What is an Image mathematically?

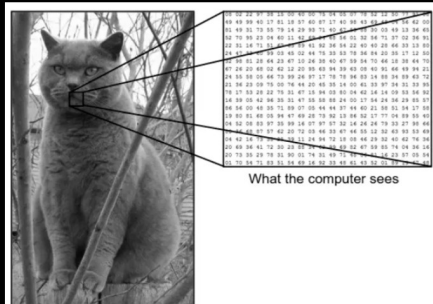
CNN Basics

Google Vision Transformer (ViT)

CLIP, ...

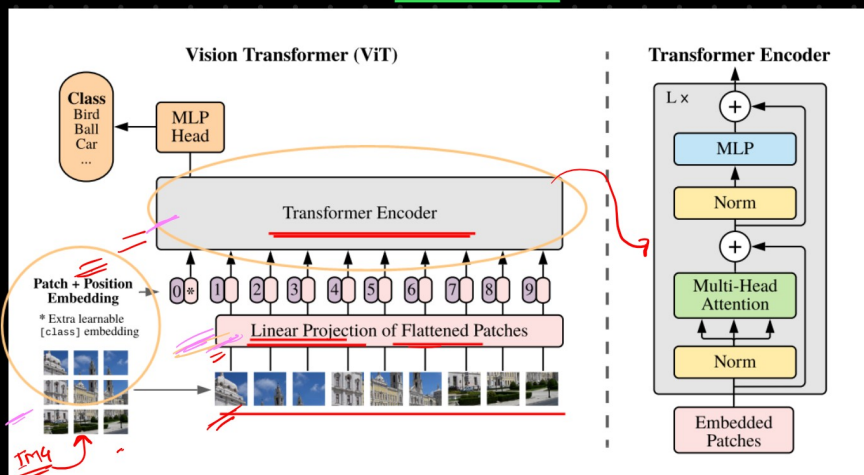
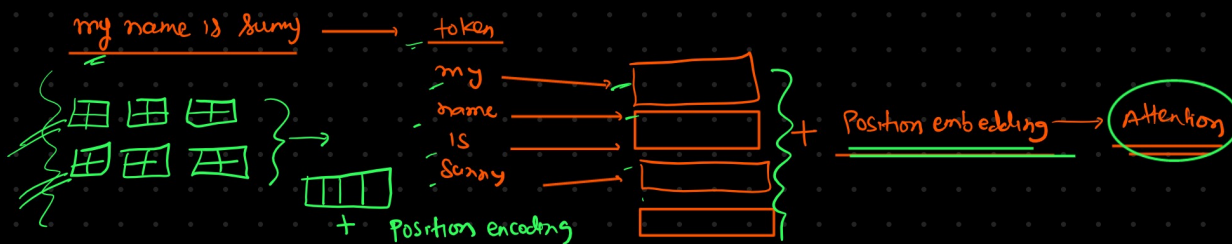
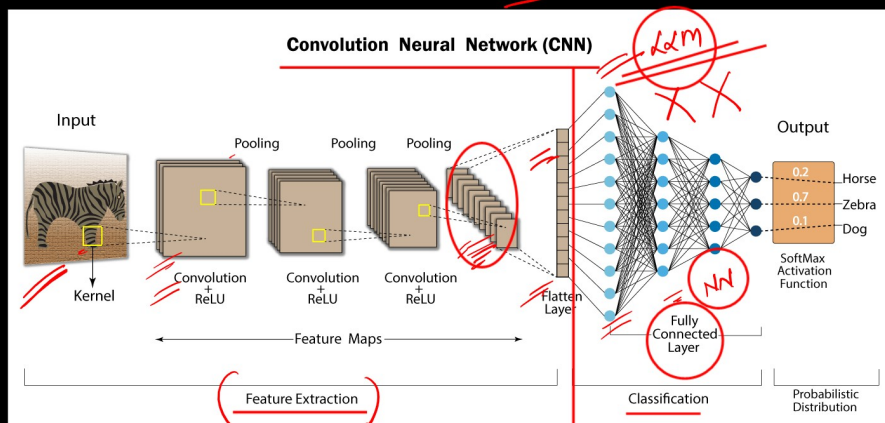
LLaVA, Qwen(image to text)

Dall-E, Imagen(Nano Banana) (text to image)



Pixel
(0-255)
BEW $\Rightarrow$ 2D
Color $\Rightarrow$ 3D
(RGB)

ViT



Image

Vision Encoder (CNN)

Image features (vector)

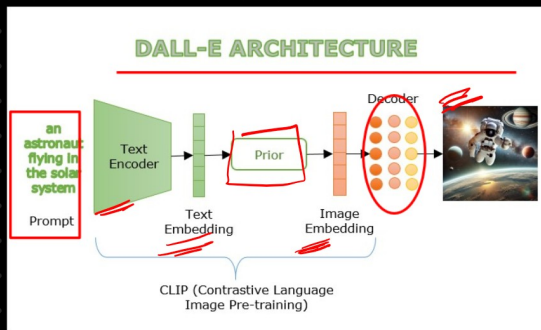
Projection Layer

Converted embeddings

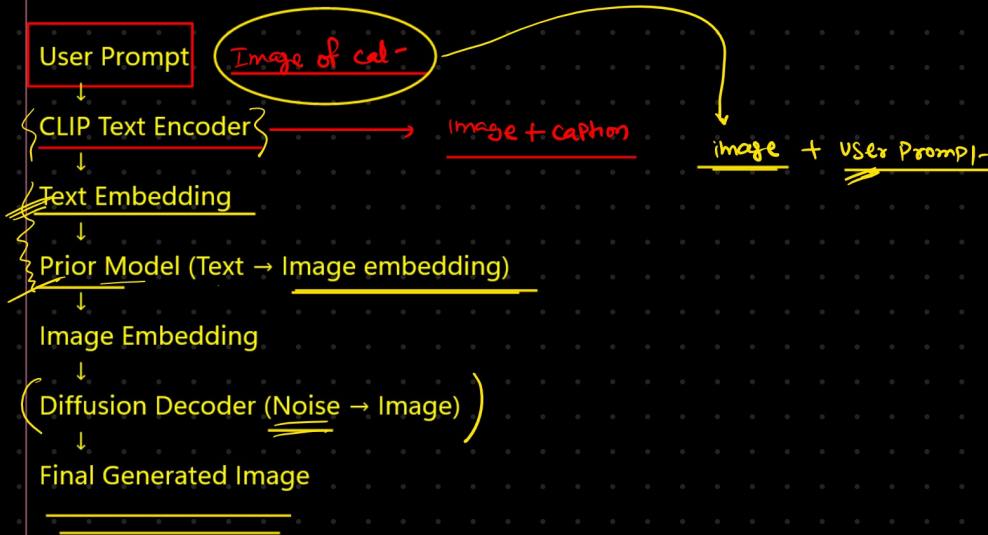
LLM

Text output

Core idea (ViT): The paper proves you can do image recognition with a pure Transformer (no CNN) by splitting an image into fixed-size patches (e.g., 16x16), flattening each patch into a vector, linearly embedding them like "tokens," adding positional embeddings, and feeding the sequence into a standard Transformer encoder



{ CLIP } embedding
 { Diffusion }



What Exactly Gets Fine-Tuned?

There are multiple strategies in multimodal fine-tuning:

Image to text

Option 1: Train Everything (Very Expensive)

Train: Vision encoder, Projection, LLM

Used by large labs with massive compute

Requires very large dataset + multi-node GPU clusters

Not practical for individual creators.

Option 2: Train Only Projection Layer (Cheapest)

Freeze: Vision encoder, LLM

Train: Projection layer (vision → LLM embedding mapper)

Used in early LLaVA (v1) style setups

Purpose: Teach LLM how to interpret visual embeddings

Limitation: Weak reasoning improvement because LLM remains fully frozen.

Option 3: Train Projection + LLM (Common)

Freeze: Vision encoder

Train: Projection layer, LLM (LoRA or full fine-tuning)

This is the most widely used approach today

Improves multimodal reasoning significantly

Most open-source multimodal models follow this.

Option 4: LoRA-based Multimodal Fine-Tuning (Practical)

✓ Vision Layers → LoRA applied

✓ LLM Layers (Attention, MLP) → LoRA applied

Means:

You are NOT fully training them.

You are training LoRA adapters inside:

This:

Works on ~24GB GPU (depending on model size)

Memory efficient

Dataset format: Dataset format is different from text-only training.

♦ Example 1: Simple Image + Instruction

```
{  
  "image": "example.png",  
  "text" or "caption": "I am sunny"  
}
```

Path | object - Image

Image to text

♦ Example 2: VQA Format

```
{  
  "image": "chart.png",  
  "question": "What is the highest value?",  
  "answer": "The highest value is 72."  
}
```

♦ Example 3: LLaVA Format

```
{  
  "image": "image_001.jpg",  
  "conversations": [  
    {"role": "user", "content": "What is in this image?"},  
    {"role": "assistant", "content": "A dog sitting on a sofa."}  
  ]  
}  
  
{  
  "id": "0001",  
  "image": "image1.jpg",  
  "conversations": [  
    {"from": "human", "value": "<image>\nDescribe the scene."},  
    {"from": "gpt", "value": "A busy street with cars and pedestrians."}  
  ]  
}
```

Note: <image> token tells model where visual embedding goes.

Aspect	Normal LLM Fine-Tuning	Multimodal LLM Fine-Tuning
Input	<u>Text only</u>	<u>Image/Audio/Video + Text</u>
Architecture	<u>LLM only</u>	<u>Encoder + Projection + LLM</u>
Compute	<u>Medium</u>	<u>Medium to High</u>
GPU Need	<u>Moderate VRAM</u>	<u>Higher VRAM (often)</u>
Complexity	<u>Medium</u>	<u>High</u>
Preprocessing	<u>Tokenization</u>	<u>Media processing + Tokenization</u>
Stability	<u>More stable</u>	<u>More sensitive</u>